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JORDAN PROPERTY FOR AUTOMORPHISM GROUPS OF COMPACT VARIETIES

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Abstract. In this note, we report some recent progress on the Jordan property for (birational) automorphism groups of projective varieties and compact complex varieties.

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1. Introduction

The story starts with a celebrated theorem proven by Camille Jordan [Jor78, Section 40] in 1878. This theorem establishes that for any field k of characteristic zero and any positive integer n, the linear automorphism group $\mathrm{GL}_n(k)$ of an n-dimensional k-vector space possesses the *Jordan property*: there exists a Jordan constant $\mathrm{J} = \mathrm{J}(n)$ such that any finite subgroup H of $\mathrm{GL}_n(k)$ contains an abelian subgroup H_1 with index $[H : H_1]$ not exceeding $\mathrm{J}(n)$.

Researchers subsequently pondered whether other automorphism groups, such as general (non-linear) automorphism groups or even birational automorphism groups of varieties, exhibit the same Jordan property.

More than a century later, we have gained a substantial understanding of the finite subgroups of the Cremona group of rank 2, denoted as $\mathrm{Cr}_2(k) = \mathrm{Bir}(\mathbb{P}^2_k)$. Notably, when the characteristic of k is zero, the work of Jean-Pierre Serre [Ser09, Theorem 5.3]

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confirms that $\mathrm{Cr}_2(k)$ possesses the Jordan property with $\mathrm{J} \leqslant 2^{10} \cdot 3^4 \cdot 5^2 \cdot 7$. Later, Popov explicitly and systematically stated in [Pop11, Pop14] several problems in the general setting involving algebraic geometry, complex geometry and differential geometry.

In this survey, we will primarily highlight recent advancements, following Serre's work. The following is an outline of the structure of this article:

In Section 2, we will introduce some terminology and notation, and state some fundamental and crucial lemmas related to the Jordan property. In Section 3, we will focus on birational automorphism groups in characteristic 0, summarizing the significant works of of Bandman, Birkar, Popov, Prokhorov, Shramov and Zarhin. In particular, we demonstrate that Serre's results have been comprehensively generalized to all dimensions. In Section 4, we will elaborate on the Jordan property for automorphism groups of projective varieties. In Section 5, we will delve into the p-Jordan and generalized p-Jordan properties in positive characteristics. In Section 6, we will present two distinct strategies for addressing Fujiki's class $\mathcal{C}$. Finally, in Section 7, we will enumerate several partial results pertaining to general compact complex manifolds.

For groups acting smoothly or continuously on a real manifold, its Jordan property has be intensively studied and we refer to Mundet i Riera [MiR19, §1] for the excellent survey of the related results.

We refer to [BZ23] and [BZ24], for two more excellent surveys on the Jordan property of (bimeromorphic) automorphism groups of holomorphic $\mathbb{P}^1$-bundles, and the interrelations between geometric properties of complex projective varieties or compact Kähler manifolds and the Jordan property, or the lack of it, of their (bimeromorphic) automorphism groups.

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2. Preliminaries

We recall and adopt some notations and easy lemmas from [MZ18] for the convenience of the readers. Most of the concepts and results are well-known and appeared in early literature with possibly different formulation.

Definition 2.1 (cf. [MZ18, Definition 2.2])**.** Given a group G, we introduce the following constants:

$$\mathrm{Bd}(G) = \sup\{|F| : |F| < \infty, F \leqslant G\},$$
$$\mathrm{Rk_f}(G) = \sup\{\mathrm{Rk_f}(A) : |A| < \infty, A \text{ is abelian}, A \leqslant G\},$$

where $\mathrm{Rk_f}(A)$ is the *minimal number* of generators of a finite abelian group A. Similarly, we may define these constants for a family of groups.

Definition 2.2. A group G (resp. a family $\mathcal{G}$ of groups) has the *Jordan property* if there exists a Jordan constant $\mathrm{J} = \mathrm{J}(G)$ (resp. $\mathrm{J} = \mathrm{J}(\mathcal{G})$) such that any finite subgroup H of G (resp. $G \in \mathcal{G}$) contains an abelian subgroup H_1 with index $[H : H_1]$ not exceeding J. We use $\mathrm{J'} = \mathrm{J'}(G)$ (resp. $\mathrm{J'} = \mathrm{J'}(\mathcal{G})$) if we assume $H_1 \lhd H$ in the above definition.

Remark 2.3. Note that $\mathrm{J}(G) \leqslant \mathrm{J'}(G) \leqslant \mathrm{J}(G)!$. Indeed, for any finite subgroup H and its abelian subgroup H_1 with $[H : H_1] \leqslant \mathrm{J}(G)$. The action of H on H/H_1 yields a homomorphism of H to the symmetric group on $[H : H_1]$ letters. The kernel of this homomorphism is a normal subgroup H_2 of H, of index dividing $\mathrm{J}(G)!$, and contained in H_1.

The easy observations below are frequently used.

Lemma 2.4 ([MZ18, Lemma 2.3])**.** *Consider the exact sequence of groups*

$$1 \to G_1 \to G \to G_2 \to 1.$$

(1) $\mathrm{J}(G) \leqslant \mathrm{Bd}(G_2) \cdot \mathrm{J}(G_1)$.
(2) *If* G_1 *is finite, then* $\mathrm{J}(G_2) \leqslant \mathrm{J}(G)$.
(3) *If* $\mathrm{Bd}(G_1) = 1$, *then* $\mathrm{J}(G) \leqslant \mathrm{J}(G_2)$.
(4) *If* $G \cong G_1 \times G_2$, *then* $\mathrm{J}(G) \leqslant \mathrm{J}(G_1) \cdot \mathrm{J}(G_2)$.
(5) $\mathrm{Rk_f}(G) \leqslant \mathrm{Rk_f}(G_1) + \mathrm{Rk_f}(G_2)$.
(6) $\mathrm{J}(G) \leqslant \mathrm{J}(G_2) \cdot \mathrm{Bd}(G_1)^{\mathrm{Rk_f}(G_2)\cdot\mathrm{Bd}(G_1)}$.

One should be careful that the extension of two Jordan groups may no longer be Jordan. This is also the key obstruction making the Jordan problem challenging. The following geometric example illustrates how the reduction of the Jordan property via geometric canonical fibrations is not automatic.

Example 2.5. *Let $X = \mathbb{P}^1 \times E$ where E is an elliptic curve. Let $G = \mathrm{Bir}(X)$. Then we have an exact sequence*

$$1 \to G_1 \to G \to G_2 \to 1$$

where the map $G \to G_2 := \mathrm{Bir}(X)|_E$ is obtained from the albanese map $X \to \mathrm{Alb}(X) = E$, $G_2 \leqslant \mathrm{Aut}(E) \cong E \rtimes F$ for some finite group F and the kernel (of $G \to G_2$) $G_1 \leqslant \mathrm{PGL}(2, k(E))$. Clearly, G_1 and G_2 are Jordan (cf. Theorem 4.1). However, G is not Jordan (cf. Theorem 3.1).

Lemma 2.6 (cf. [MZ18, Lemma 2.4])**.** *Below are some important constants.*

(1) *(Jordan) Let $\mathrm{C}_n := \mathrm{J'}(\mathrm{GL}_n(k))$ (cf. Definition2.2,* [CR62, Theorem 36.14, pp. 258-262]*). Every general linear group $\mathrm{GL}_n(k)$ is a Jordan group, so every linear algebraic group is a Jordan group. We have $\mathrm{C}_n = (n+1)!$ afforded by S_{n+1} when $n \geqslant 71$; see* [Col07] *for the exact value of C_n.*

(2) *(Minkowski) $\mathrm{Bd}(\mathrm{GL}_n(K)) < \infty$ for any field K which is finitely generated over $\mathbb{Q}$. The bound depends on n and $[K : \mathbb{Q}]$ (cf.* [Ser07, Theorem 5, and §4.3]*).*

(3) $\mathrm{Rk_f}(T) = \dim T$*, when T is an algebraic torus.*

(4) $\mathrm{Rk_f}(\mathrm{GL}_n(k)) = n$ *(see* [Hu75, §15.4 Proposition] *and use (3)).*

(5) $\mathrm{Rk_f}(A) = 2 \dim A$*, when A is an abelian variety.*

In this survey, we will mainly focus on automorphism groups and birational automorphism groups of varieties (mostly projective or compact).

Notation 2.7. Let X be an algebraic variety. Denote by $\mathrm{Aut}(X)$ the group of all automorphisms of X and $\mathrm{Bir}(X)$ the group of all birational self-maps of X. When X is a complex space, $\mathrm{Aut}(X)$ (resp. $\mathrm{Bir}(X)$) refers to the group of all biholomorphic (resp. bimeromorphic) self-maps of X.

Given an $\mathbb{R}$ -Cartier divisor L in a projective variety X, denote by

$$\mathrm{Aut}_{[L]}(X) := \{g \in \mathrm{Aut}(X) \,|\, g^*L \equiv L\}$$

where $g^*L \equiv L$ means numerical equivalence. Given a class $[\alpha] \in H^{1,1}(X, \mathbb{R})$ on a compact complex manifold X, denote by

$$\mathrm{Aut}_{[\alpha]}(X) := \{g \in \mathrm{Aut}(X) \,|\, g^*[\alpha] = [\alpha]\}.$$

By using the Hilbert Scheme when X is a projective variety (cf. [Kol96]), we have the following natural inclusion

$$\mathrm{Aut}(X) \subseteq \mathrm{Bir}(X) \subseteq \mathrm{Hilb}(X \times X)$$

via $g \in \mathrm{Bir}(X) \mapsto \Gamma_g \in \mathrm{Hilb}(X \times X)$ where Γ_g is the graph of g. Note that both $\mathrm{Aut}(X)$ and $\mathrm{Bir}(X)$ are open subschemes of $\mathrm{Hilb}(X \times X)$. This way, we denote by $\mathrm{Aut}_0(X)$ the

neutral component of $\mathrm{Aut}(X)$. A similar statement works by using the Douady space when X is a compact complex variety, see [Fuj78].

However, $\mathrm{Hilb}(X \times X)$ and also $\mathrm{Aut}(X)$ may have infinitely many connected components. So $\mathrm{Aut}(X)$ may not be a variety. Such an example is easy to construct whenever X admits an automorphism of positive entropy (e.g. when X is a product of elliptic curves). Nevertheless, if we fix a Hilbert polynomial P of a subvariety, such as the diagonal of $X \times X$ (the graph of the identity map), then $\mathrm{Aut}(X) \cap \mathrm{Hilb}_P(X \times X)$ will be a group variety with only finitely many connected components. Here, $\mathrm{Hilb}_P(X \times X) \subseteq \mathrm{Hilb}(X \times X)$ is the subscheme of subvarieties with Hilbert polynomial P, and it is a projective variety (cf. [Kol96, Theorem 1.4]). In this way, we obtain the following significant lemma for the reduction to the connected group.

Lemma 2.8 ([MZ18, Lemma 2.5])**.** *Let X be a projective variety. Then there exists a constant ℓ (depending on X), such that for any finite subgroup $G \leqslant \mathrm{Aut}(X)$, we have $[G : G \cap \mathrm{Aut}_0(X)] \leqslant \ell$.*

3. Birational automorphism groups in characteristic 0

In this section, we always work over fields of characteristic 0 unless otherwise specified. It is Popov who asked whether the group $\mathrm{Aut}(X)$ (resp. $\mathrm{Bir}(X)$) of all automorphisms (resp. all birational automorphisms) of an algebraic variety X is Jordan (cf. [Pop11, Question 2.30-2.31]). Popov [Pop14] predicted that $\mathrm{Aut}(X)$ should be Jordan and till now there is no known counter example. We shall see in the next section, this has been verified when X is projective. In this section, we focus on $\mathrm{Bir}(X)$.

Popov himself proved that for a projective surface X, the group $\mathrm{Bir}(X)$ is Jordan unless X is birational to the product $\mathbb{P}^1 \times E$ with E being an elliptic curve. Later, Zarhin confirmed that $\mathrm{Bir}(\mathbb{P}^1 \times E)$ is not Jordan (cf. [Pop11, §2.2], [Ser09, Theorem 5.3], [Zar14, Theorem 1.2], [Zar15, Theorem 1.3]). In dimension three, Prokhorov and Shramov [PS18, Theorem 1.8] gave fully classification of threefolds with non-Jordan birational automorphism groups; see also [PS20] for the case of uniruled Kähler threefolds. We summarize as follows.

Theorem 3.1. *Let X be an algebraic surface. Then $\mathrm{Bir}(X)$ is Jordan if and only if X is not birational to $\mathbb{P}^1 \times E$ with E an elliptic curve.*

Theorem 3.2. *Let X be an algebraic threefold. Then $\mathrm{Bir}(X)$ is Jordan if and only if X is neither birational to $E \times \mathbb{P}^2$ with E an elliptic curve, nor to $S \times \mathbb{P}^1$ with S being one of the following:*

- *an abelian surface;*

- *a bielliptic surface;*
- *a surface of Kodaira dimension 1 such that the Jacobian fibration of the pluri-canonical fibration is Zariski locally trivial.*

Remark 3.3. If one considers the solvably (resp. nilpotently) Jordan property which replaces the abelian subgroup by the solvably (resp. nilpotent) subgroup, then for complex algebraic variety X, the group $\mathrm{Bir}(X)$ is always solvably (resp. nilpotently) Jordan (cf. [PS14, PS16, Gul20]).

In higher dimensions, with the help of the minimal model program (cf. [BCHM10]), Prokhorov and Shramov [PS14, Theorem 1.8] (see also [PS16, Theorem 1.8]) confirmed the Jordan property of the group $\mathrm{Bir}(X)$ for any algebraic variety X, assuming that either X is non-uniruled or X has vanishing irregularity as well as the (then) outstanding Borisov-Alexeev-Borisov conjecture about the boundedness of terminal Fano varieties which has now been affirmatively confirmed by Birkar [Bir21, Theorem 1.1]. In particular, the result applies when X is rationally connected (i.e., any two points are connected by a rational curve).

We list these results as follows and sketch their ideas. Recall that, for an algebraic variety X, we define

$$\widetilde{q}(X) = h^1(X', \mathcal{O}_{X'})$$

where X' is any smooth projective birational model of X.

Theorem 3.4 ([PS14, Theorem 1.8] and [Bir21, Corollary 1.5])**.** *Let X be a rationally connected algebraic variety. Then* $\mathrm{Bir}(X)$ *is Jordan with the Jordan constant depending only on* $\dim X$.

To prove Theorem 3.4, it suffices to show that for any finite group G (faithfully) acting on a terminal rationally connected variety X, there is a subgroup $H \leqslant G$ with index $[G : H] \leqslant C$, where C depends only on $\dim X$, such that H has a fixed point on X. Such idea was first introduced by Popov in [Pop14]. Further reduction argument allows us to focus only on terminal Fano X. This way, there exists some (minimal) $m > 0$ such that the anti-pluri-canonical divisor $-mK_X$ is a very ample Cartier divisor and defines an G-equivariant embedding $X \hookrightarrow \mathbb{P}^N$ where $N = h^0(X, -mK_X) - 1$. So $G \leqslant \mathrm{PGL}(N+1)$. Finally, the Borisov-Alexeev-Borisov conjecture, proved by Birkar, tells that such N is upper bounded by a function in $\dim X$.

Remark 3.5. In particular, all Cremona groups $\mathrm{Cr}_n(k) = \mathrm{Bir}(\mathbb{P}^n_k)$ have the Jordan property, confirming a conjecture of Serre. Here, we list some known results for the Jordan constants.

(1) Yasinsky [Yas17, Theorems 1.9 and 1.10] showed that the normal Jordan constant $\mathrm{J'}(\mathrm{Cr}_2(k)) = 7200$ when k is algebraically closed of characteristic 0 and $\mathrm{J'}(\mathrm{Cr}_2(\mathbb{R})) = \mathrm{J'}(\mathrm{Cr}_2(\mathbb{Q})) = 120$.
(2) Zaitsev [Zai24, Theorem 1.3 and Corollary 1.4] fully classified $\mathrm{J'}(\mathrm{Cr}_2(k))$ for any k of characteristic 0 and there are exactly three values $7200, 168, 120$.
(3) Prokhorov and Shramov [PS23] showed that $\mathrm{J'}(\mathrm{Cr}_2(\mathbb{F}_q)) = |\operatorname{PGL}_3(\mathbb{F}_q)| = q^3 \cdot (q^2-1) \cdot (q^3-1)$ when $q \notin \{2, 4, 8\}$. Later, Vikulova [Vik23] showed that this is true when $q \neq 2$ and the only exceptional case is $\mathrm{J'}(\mathrm{Cr}_2(\mathbb{F}_2)) = |S_6| = 720 > |\operatorname{PGL}_3(\mathbb{F}_2)|$.
(4) Prokhorov and Shramov [PS17, Theorem 1.2.4] gave a uniform upper bound $\mathrm{J'}(\mathrm{Bir}(X)) \leqslant 107495424$ for any rationally connected threefold X.

We reorganize the strategy in [PS14] and provide a slightly streamlined proof for the following theorem.

Theorem 3.6 (cf. [PS14, Theorem 1.8])**.** *Let X be a non-uniruled variety. Then* $\mathrm{Bir}(X)$ *is Jordan.*

Proof. A projective variety Y is *quasi-minimal* if Y has terminal singularities and the canonical divisor K_Y is movable (cf. [PS14, Definition 4.2]). Note that X admits a quasi-minimal model by simply running a minimal model program till no divisorial contraction occurs; see the last part of the proof of [PS14, Lemma 4.4] and take the group Γ there to be the trivial group. So far, we do not need to consider any finite group action or run any equivariant minimal model program as in [PS14, Lemma 4.4] and its proof.

Now we may assume that X is quasi-minimal and $\mathbb{Q}$-factorial. Denote by

$$\mathrm{PAut}(X) := \{g \in \mathrm{Bir}(X) \mid g \text{ is isomorphic in codimension one}\}$$

the group of pseudo-automorphisms. It is known that $\mathrm{Bir}(X) = \mathrm{PAut}(X)$, see [PS14, Proposition 4.6]. Consider the exact sequence

$$(*) \quad 1 \to G \to \mathrm{PAut}(X) \to \mathrm{PAut}(X)|_{\mathrm{NS}(X)} \to 1$$

where the restriction $\mathrm{PAut}(X)|_{\mathrm{NS}(X)}$ is the image of $\mathrm{PAut}(X) \to \mathrm{GL}(\mathrm{NS}(X))$ induced by the pullback action, and it makes sense because the small birational maps do not affect divisors. Now the key observation is that $G \leqslant \mathrm{Aut}(X)$ by noting that any $g \in G$ satisfies $g^*A \equiv A$ for an ample divisor A and the negativity lemma, see [JM24a, Proposition 2.1] for a more generalized setting. Here, we provide a simple algebraic version in Lemma 3.7.

We shall see that $\mathrm{Aut}(X)$ is Jordan by Theorem 4.1. So the theorem is proved by noting that $\mathrm{PAut}(X)|_{\mathrm{NS}(X)} \leqslant \mathrm{GL}(\mathrm{NS}(X))$ has bounded finite subgroups. □

Lemma 3.7. *Let $f \in \mathrm{PAut}(X)$ with X being a $\mathbb{Q}$-factorial normal projective variety over an algebraically closed field k of arbitrary characteristic. Suppose $f^*A \equiv A$ for some ample divisor A. Then $f \in \mathrm{Aut}(X)$.*

Proof. Let Y be the normalization of the graph of f. Denote by $p_i : Y \to X$ the two induced projections for $i = 1, 2$. Since f is isomorphic in codimension 1, p_1 and p_2 share the same exceptional divisors. Note that p_i are birational and $f^*A = {p_1}_* p_2^* A \equiv A$.

We claim that ${p_1}^* f^* A = p_2^* A$. Note that $F := p_1^* {p_1}_* p_2^* A - p_2^* A$ is an effective p_1-exceptional (and hence p_2-exceptional) divisor. By the assumption, $F \equiv p_1^* A - p_2^* A$ is p_2-nef. By the negativity lemma (cf. [KM98, Lemma 3.39]), $-F$ is effective. So $F = 0$ and the claim is proved.

Let $C \subseteq Y$ be any curve such that $p_1(C)$ is a point. Then

$$A \cdot {p_2}_* C = p_2^* A \cdot C = p_1^* f^* A \cdot C = 0$$

by the projection formula. Therefore, $p_2(C)$ is a point. By the rigidity lemma (cf. [Deb01, Lemma 1.15]), f is well-defined. Similarly, f^{-1} is well-defined. □

Remark 3.8. In the proof of Theorem 3.6, we actually have $G \leqslant \mathrm{Aut}_{[L]}(X)$ (a finite extension of $\mathrm{Aut}_0(X)$ by a classical result due to Fujiki [Fuj78] and Lieberman [Lie78] independently) for any ample divisor L. If the quasi-minimal X further has $\widetilde{q}(X) = 0$, then $\mathrm{Aut}_0(X)$ is trivial by [Fuj78] and [Lie78] and hence G is finite. In particular, $\mathrm{Bir}(X)$ has bounded finite subgroups (i.e., $\mathrm{Bd}(\mathrm{Bir}(X)) < \infty$) when X is non-uniruled and $\widetilde{q}(X) = 0$.

In general, we make use of the maximal rationally connected fibration $\varphi : X \dashrightarrow Y$ where Y is non-uniruled and the geometric generic fibre $X_{\overline{k(Y)}}$ of φ is rationally connected. Note that $\mathrm{Bir}(X)$ descends to Y and we have the following exact sequence

$$1 \to G \to \mathrm{Bir}(X) \to \mathrm{Bir}(X)|_Y \to 1$$

where $\mathrm{Bir}(X)|_Y$ is the image of $\mathrm{Bir}(X) \to \mathrm{Bir}(Y)$ and $G \leqslant \mathrm{Bir}(X_{\overline{k(Y)}})$ is Jordan by Theorem 3.4. Suppose further $\widetilde{q}(X) = 0$. Then $\widetilde{q}(Y) = 0$ and hence $\mathrm{Bir}(X)|_Y \leqslant \mathrm{Bir}(Y)$ has bounded finite subgroups by Remark 3.8. Now Lemma 2.4 implies the following:

Theorem 3.9 (cf. [PS14, Theorem 1.8])**.** *Let X be an algebraic variety with $\widetilde{q}(X) = 0$. Then $\mathrm{Bir}(X)$ is Jordan.*

Remark 3.10. By Lemma 2.4, Rk_{f} is subadditive under group extension.

Thus one can show that $\mathrm{Rk}_{\mathrm{f}}(\mathrm{Bir}(X)) < \infty$ for any algebraic variety by the previous discussion.

Indeed, via the exact sequence induced by the maximal rationally connected fibration, it suffices to show $\mathrm{Rk_f}(\mathrm{Bir}(X)) < \infty$ when X is quasi-minimal and when X is rationally connected.

When X is quasi-minimal, $\mathrm{Rk_f}(\mathrm{Aut}(X)) < \infty$ and $\mathrm{Rk_f}(\mathrm{PAut}(X)|_{\mathrm{NS}(X)}) < \infty$ by Theorem 4.1 and Lemma 2.6, so $\mathrm{PAut}(X)$ (and $\mathrm{Aut}(X)$) have $\mathrm{Rk_f}$ being finite by the exact sequence $(*)$ in Theorem 3.6 and the subadditivity of $\mathrm{Rk_f}$.

When X is rationally connected, by the same G-MMP strategy as in the proof of [PS14, Lemma 7.6], it suffices to show $\mathrm{Rk_f}(\mathrm{Bir}(X)) < \infty$ when X is terminal Fano. As we have discussed, there exists some $N(n) > 0$ such that for any finite group $G \leqslant \mathrm{Aut}(X)$ with X being n-dimensional terminal Fano, $G \leqslant \mathrm{PGL}_{N(n)}(k)$. So $\mathrm{Rk_f}(\mathrm{Bir}(X)) \leqslant \mathrm{Rk_f}(\mathrm{PGL}_{N(n)}(k)) < \infty$ (cf. Lemma 2.6). Tracing the argument, one sees that $\mathrm{Rk_f}(\mathrm{Bir}(X))$ is bounded for all n-dimensional rationally connected X.

We shall see the progress on positive characteristics in Section 5.

4. Automorphism groups in characteristic 0

In this section, we always work over fields of characteristic 0. The second and third authors proved that $\mathrm{Aut}(X)$ is Jordan when X is a projective variety, which answers the projective case of [Pop14, §2, Problem A].

Theorem 4.1 ([MZ18, Theorem 1.6]). *Let X be a projective variety. Then $\mathrm{Aut}(X)$ is a Jordan group and $\mathrm{Rk_f}(\mathrm{Aut}(X)) < \infty$.*

The approach in [MZ18] towards Jordan property for the full automorphism group $\mathrm{Aut}(X)$ of a projective variety X in arbitrary dimension is more algebraic-group theoretical. It does not use the classification of projective varieties. To be specific, the authors started by proving that every (not necessarily linear) algebraic group has the Jordan property, which is a direct corollary of either one of the following two stronger results.

Let $\mathcal{G}$ be a family of groups. We say $\mathcal{G}$ is *uniformly Jordan* if $\{J(G)\}_{G\in\mathcal{G}}$ is upper bounded.

Theorem 4.2 ([MZ18, Theorem 1.3]). *Fix an integer $n \geqslant 0$. Let $\mathcal{G}$ be the family of all connected algebraic groups of dimension n. Then $\mathcal{G}$ is uniformly Jordan and $\mathrm{Rk_f}(\mathcal{G}) < \infty$.*

Theorem 4.3 ([MZ18, Theorem 1.4]). *Fix an integer $n \geqslant 0$. Let $\mathcal{G} = \{\mathrm{Aut}_0(X) \,|\, X$ is a projective variety of dimension $n\}$. Then $\mathcal{G}$ is uniformly Jordan and $\mathrm{Rk_f}(\mathcal{G}) < \infty$.*

Remark 4.4. Theorem 4.3 cannot be deduced directly from Theorem 4.2 because the dimension of $\mathrm{Aut}_0(X)$ cannot be bounded solely in terms of $\dim X$. For instance, according to [Mar71, Theorem 3], if $\mathbb{F}_d$ denotes the Hirzebruch surface of degree $d \geqslant 1$, then

the unipotent radical of $\mathrm{Aut}_0(\mathbb{F}_d)$ is $\mathbb{G}_a^{d+1}$, where $\mathbb{G}_a = (k, +)$ represents the additive group. Notably, $\dim \mathrm{Aut}_0(\mathbb{F}_d) = d + 5$, which is not bounded (in terms of $\dim \mathbb{F}_d = 2$). To establish Theorem 4.3, we mitigate the impact of such unipotent radicals through a crucial lemma: [MZ18, Lemma 3.7].

Remark 4.5. We briefly outline the strategy of the proofs for the aforementioned two theorems. Let G be a connected algebraic group, and let G_{aff} represent the largest connected affine normal subgroup. It is important to note that any anti-affine subgroup of G is connected and lies within the centre of G, thereby being normal in G. We denote the largest anti-affine subgroup by G_{ant}. The foundation of our proof lies in the classic decomposition theorem $G = G_{\mathrm{aff}} \cdot G_{\mathrm{ant}}$ for a connected algebraic group G. This theorem is attributed to Rosenlicht [Ros56, Corollary 5, p. 440], and further modern elaborations can be found in Brion [Bri09]. We also use the effective (and optimal) upper bound in Brion [Bri13, Proposition 3.2] for the dimension of the anti-affine part of $\mathrm{Aut}_0(X)$. For any projective variety X and its normalization X', $\mathrm{Aut}(X)$ lifts to X'. So it suffices to prove Theorem 4.1 when X is normal. Then, Lemma 2.8, Theorem 4.2 and an application of [MZ18, Lemma 3.7] imply Theorem 4.1.

At the end of this section, we list two well known related open questions. We are only interested in the situation when $\mathrm{Bir}(X)$ is not Jordan, see Theorem 3.9. The following question was first formulated by Popov in [Pop11, Corollary 2.30].

Question 4.6. *Let X be an algebraic variety. Is $\mathrm{Aut}(X)$ Jordan?*

Bandman and Zarhin proved that $\mathrm{Aut}(X)$ is Jordan when $\dim X = 2$ or X is birational to the product $\mathbb{P}^1 \times A$ with A a smooth projective variety having no rational curve; see [BZ15, Theorem 1.7] and [BZ19, Theorem 4].

Question 4.7. *Let X be a projective variety and $r > 0$. Let $\mathcal{G}$ be the family of $\mathrm{Aut}(X')$ with X' birational to X and the Picard number $\rho(X') \leqslant r$. Is $\mathcal{G}$ uniformly Jordan?*

Bounding the Picard number is necessary by noting that $\mathrm{Bir}(\mathbb{P}^1 \times E)$ is not Jordan as in Theorem 3.1, and its finite subgroup (action) can always be regularized on some high birational model of $\mathbb{P}^1 \times E$. A natural strategy is to apply Lemma 2.8. However, we do not know whether the ℓ there depends only on the Picard number.

5. Positive characteristic

The Jordan property for $\mathrm{GL}_n(k)$ does not hold in general when $\mathrm{char}\, k = p > 0$ due to the existence of unipotent elements of finite order. For instance, the group $\mathrm{GL}_n(\overline{F}_p)$ contains arbitrarily large subgroups of the form $\mathrm{SL}_n(F_{p^r})$ which are simple modulo their

centres. Nevertheless, for any finite subgroup Γ of $\mathrm{GL}_n(k)$ of order not divisible by p, there still exists a normal abelian subgroup A of Γ with $[\Gamma : A] \leqslant J(n)$ for some constant $J(n)$. Later, Serre [Ser09, Theorem 5.3] showed that the Cremona group $\mathrm{Cr}_2(k)$ of rank 2 over a field k also has this property. This motivates the following definition.

Definition 5.1 ([Hu20, Definition 1.2])**.** Let p be a prime number or zero. A group G is called a *generalized p-Jordan group* if there exists a constant $J(G)$, depending only on G, such that every finite subgroup Γ of G whose order is not divisible by p contains a normal abelian subgroup A of index $\leqslant J(G)$. We also call such a group generalized Jordan if there is no confusion caused (e.g., in practice, p will always denote the characteristic of the ground field k).

Note that when $p = 0$, this notion coincides with Popov's [Pop11, Definition 2.1]. The above-mentioned results can be reformulated as follows. Both general linear groups $\mathrm{GL}_n(k)$ and the Cremona group $\mathrm{Cr}_2(k)$ of rank 2 are generalized p-Jordan, where $p = \mathrm{char}(k)$. It is proved in [Hu20] that, in addition to the above, any algebraic group is generalized Jordan.

Theorem 5.2 ([Hu20, Theorem 1.3])**.** *Any algebraic group G defined over a field k of characteristic $p \geqslant 0$ is generalized p-Jordan. Namely, there exists a constant $J(G)$, depending only on G, such that every finite subgroup Γ of $G(k)$ whose order is not divisible by p contains a normal abelian subgroup of index $\leqslant J(G)$.*

Another generalization of Jordan's theorem to prime characteristic was due to Brauer and Feit [BF66] by allowing an arbitrary finite subgroup Γ of $\mathrm{GL}_n(k)$ whose order may be divisible by $p > 0$. They showed that Γ contains a normal abelian subgroup whose index is bounded by a constant depending on n as well as the order of the p-Sylow subgroup $\Gamma^{(p)}$ of Γ. Larsen and Pink subsequently extended Brauer–Feit [BF66] as follows.

Theorem 5.3 ([LP11, Theorem 0.4])**.** *For any positive integer n, there exists a constant $J'(n)$ such that any finite subgroup Γ of $\mathrm{GL}_n(k)$ over a field k of characteristic $p > 0$ contains a normal abelian p'-subgroup A of index $\leqslant J'(n) \cdot |\Gamma^{(p)}|^3$.*

Here a finite group is called a p-group (resp. p'-group) if its order is some power of p (resp. relatively prime to p). In an analogous way, Hu [Hu20] introduced the following notion.

Definition 5.4 ([Hu20, Definition 1.6])**.** Let p be a prime number. A group G is called a *p-Jordan group* if there exist constants $J'(G)$ and $e(G)$, depending only on G, such that every finite subgroup Γ of G contains a normal abelian p'-subgroup A of index $\leqslant J'(G) \cdot |\Gamma^{(p)}|^{e(G)}$.

The following result of Hu extends Theorem 5.3 to arbitrary algebraic groups.

Theorem 5.5 ([Hu20, Theorem 1.7]). *Any algebraic group G defined over a field k of characteristic $p > 0$ is p-Jordan. That is, there are constants $J'(G)$ and $e(G)$, depending only on G, such that any finite subgroup Γ of $G(k)$ contains a normal abelian p'-subgroup A of index $\leqslant J'(G) \cdot |\Gamma^{(p)}|^{e(G)}$.*

Together with Lemma 2.8, Hu has also deduced two Jordan-type properties for automorphism groups of projective varieties of arbitrary characteristic.

Theorem 5.6 ([Hu20, Theorems 1.9 and 1.10]). *Let X be a projective variety defined over a field k of characteristic $p \geqslant 0$. Then $\mathrm{Aut}(X)$ is p-Jordan and generalized p-Jordan.*

For birational automorphism groups, an analogue of [Ser09, Theorem 5.3] is proved by Chen and Shramov.

Theorem 5.7 ([CS21, Theorem 1.6]). *There exists a constant J such that for every prime p and every field k of characteristic p, every finite subgroup G of the birational automorphism group $\mathrm{Bir}(\mathbb{P}^2)$ contains a normal abelian subgroup of order coprime to p and index at most $J \cdot |G_p|^3$, where G_p is a p-Sylow subgroup of G. In particular, for every field k of characteristic $p > 0$, the group $\mathrm{Bir}(\mathbb{P}^2)$ is p-Jordan.*

Furthermore, in [CS21] an analog of [Pop11, Theorem 2.32] (see also [PS21, Theorem 1.7]) is proved.

Theorem 5.8 ([CS21, Theorem 1.7]). *Let k be an algebraically closed field of characteristic $p > 0$, and let S be an irreducible algebraic surface over k. Then the following assertions hold.*

(1) *If S is birational to a product $E \times \mathbb{P}^1$ for some elliptic curve E, then the group $\mathrm{Bir}(S)$ is not generalized p-Jordan.*
(2) *If the Kodaira dimension of S is negative but S is not birational to a product $E \times \mathbb{P}^1$ for any elliptic curve E, then the group $\mathrm{Bir}(S)$ is p-Jordan but not Jordan.*
(3) *If the Kodaira dimension of S is non-negative, then the group $\mathrm{Bir}(S)$ is Jordan.*

In the proof of Theorem 5.8, the following assertion is also proved, which makes Hu's result [Hu20] more precise in one important particular case.

Proposition 5.9 ([CS21, Proposition 1.8]). *Let k be an arbitrary field, and let X be a smooth geometrically irreducible projective variety of non-negative Kodaira dimension over k. Then the group $\mathrm{Aut}(X)$ is Jordan.*

6. Automorphism groups of Kähler manifolds and Fujiki's class $\mathcal{C}$

The result and strategy of [MZ18] are extended to the Kähler case by Kim [Kim18], while Popov offered a much simpler proof by reducing the Jordan property to (connected real) Lie groups. Note that Boothby and Wang [BW65] proved the Jordan property for connected real Lie groups. Popov [Pop18, Theorem 2] generalized it to any real Lie group G with $\mathrm{Bd}(G/G_0) < \infty$.

Theorem 6.1. ([BW65, Pop18]) *A real Lie group G is Jordan when $\mathrm{Bd}(G/G_0) < \infty$ where G_0 is the neutral component of G.*

Recall that a group G is called *strongly Jordan* (cf. [MiRT15]) if it is Jordan, and there exists a constant m depending only on G, such that any finite abelian subgroup A of G has rank at most m.

Theorem 6.2. ([Kim18, Theorem 1.1], [Pop18, Theorem 2]) *Let X be a compact Kähler space. Then $\mathrm{Aut}(X)$ is strongly Jordan.*

People then wonder whether the same conclusion holds for manifolds similar to projective or Kähler ones. We recall some definitions.

A compact complex manifold X is in *Fujiki's class* $\mathcal{C}$, if one of the following three equivalent conditions is satisfied:

(1) X is the meromorphic image of a compact Kähler manifold.
(2) X is bimeromorphic to a compact Kähler manifold.
(3) X admits a big $(1,1)$-class $[\alpha]$.

For the equivalence of the definitions and fundamental properties of Fujiki's class $\mathcal{C}$, we refer to [Fuj78, Definition 1.1 and Lemma 1.1], [Var89, Chapter IV, Theorem 5] and [DP04, Theorem 0.7].

A compact complex manifold X is *Moishezon* if X is bimeromorphic to a projective manifold. Hence a Moishezon manifold is in Fujiki's class $\mathcal{C}$. A Kähler Moishezon manifold is projective.

Prokhorov-Shramov [PS19] generalised Theorem 6.2 to Moishezon threefolds. Their idea is to reduce the problem to the projective case. Roughly speaking, they utilise the maximal rational connected fibration $X \dashrightarrow V$ which still exists for their Moishezon threefolds (and indeed, for all Moishezon manifolds), and the famous non-uniruled-ness of V due to Graber, Harris and Starr, to either show that X is indeed a projective variety and then apply Theorem 4.1 or show that X is a rationally connected variety and then apply [PS14], or reduce to a very general fibre (a complex surface or curve) and then apply [PS21].

Perroni and the last two authors fully proved the Fujiki's case.

Theorem 6.3 ([MPZ22, Theorem 1.1]). *Let X be a connected complex manifold and let $Z \subseteq X$ be a non-empty compact complex subspace in Fujiki's class $\mathcal{C}$. Then the automorphism group $\mathrm{Aut}(X, Z)$ of all biholomorphic automorphisms of X preserving Z is Jordan.*

The approach to Theorem 6.3 is based on a simple idea: make use of the non-Kähler locus of a big $(1,1)$ class $[\alpha]$ on an "invariant" subspace Z (in Fujiki's class $\mathcal{C}$)

$$E_{nK}(\alpha) := E_{nK}([\alpha]) := \bigcap_{T \in [\alpha]} \mathrm{Sing}(T)$$

to find some "invariant" Kähler submanifold $Z_1 \subseteq Z \subseteq X$. Here, the intersection ranges over all Kähler currents $T = \alpha + i\partial\bar{\partial}\varphi$ in the class $[\alpha]$, and $\mathrm{Sing}(T)$ is the complement of the set of points $z \in Z$ such that φ is smooth near z. Note that $E_{nK}(\alpha)$ is an analytically closed subvariety; see [Tos18] and [Bou04] for details. To maintain the "invariant" property, we may occasionally need to slightly reduce the automorphism group, while the Jordan property remains unaffected. Next, we focus on the (linear) automorphism group of the normal bundle $\mathcal{N}_{Z_1/X}$ as inspired by Mundet i Riera [MiR19]. By employing an equivariant compactification of $\mathcal{N}_{Z_1/X}$, we are able to narrow down the inquiry regarding the Jordan property to the scenario of compact Kähler manifolds.

The following result is immediately obtained by taking an equivariant resolution to reduce to the smooth case and then applying Theorem 6.3 with $Z = X$. In particular, it answers the question for the Moishezon manifolds by Prokhorov and Shramov (cf. [PS19]).

Corollary 6.4 ([MPZ22, Corollary 1.2]). *Let X be a reduced compact complex space. Then $\mathrm{Aut}(X)$ is strongly Jordan in the following cases (where (1) is a special case of (2)):*

(1) *X is Moishezon, i.e., X is bimeromorphic to a projective variety.*
(2) *X is in Fujiki's class $\mathcal{C}$, i.e., X is the meromorphic image of a compact Kähler manifold.*

Instead of studying the above invariant subgroup, a more natural way is to study the following question:

Question 6.5. *Let X be a compact complex manifold. If X is Moishezon (or in Fujiki's class $\mathcal{C}$), can one find a bimeromorphic model $\widetilde{X}$ of X such that $\mathrm{Aut}(X)$ lifts to $\widetilde{X}$, and $\widetilde{X}$ is projective (or Kähler)?*

Indeed, Demailly and Paun showed in the proof of [DP04, Theorem 3.4] that if a compact complex manifold X admits a big $(1,1)$-class $[\alpha]$, then there is a bimeromorphic holomorphic map $\sigma : X' \to X$ from a Kähler manifold X' obtained by a sequence of

blowups along smooth centres determined by the ideal sheaf J corresponding to some Kähler current $T \in [\alpha]$ with analytic singularities (cf. [DP04, Definition 2.2]).

Generally, the group $\mathrm{Aut}(X)$ may fail to lift via σ because the class $[\alpha]$ is not necessarily preserved by $\mathrm{Aut}(X)$. Consequently, we focus on the subgroup $\mathrm{Aut}_{[\alpha]}(X)$; see Notation 2.7. However, one still cannot expect the lifting of $\mathrm{Aut}_{[\alpha]}(X)$, because the blown-up ideal sheaf J may not be $\mathrm{Aut}_{[\alpha]}(X)$-invariant. Therefore, we need to find another Kähler model in a more natural way. The idea is to consider the ideal sheaf generated by g^*J for all $g \in \mathrm{Aut}_{[\alpha]}(X)$ and then we are left to verify that the new model admits a Kähler form. In this way, Jia and the second author proved the following:

Theorem 6.6 (cf. [JM24, Theorem 1.1])**.** *Let X be a compact complex manifold in Fujiki's class $\mathcal{C}$. Then, for any big $(1,1)$-class $[\alpha]$ on X, there exists a bimeromorphic holomorphic map $\sigma : X_e \to X$ from a Kähler manifold X_e such that $\mathrm{Aut}_{[\alpha]}(X)$ lifts holomorphically via σ.*

When X is a Kähler manifold with a Kähler form α, Fujiki [Fuj78, Theorem 4.8] and Lieberman [Lie78, Proposition 2.2], separately, proved that

$$[\mathrm{Aut}_{[\alpha]}(X) : \mathrm{Aut}_0(X)] < \infty.$$

Their proof heavily relies on the Kähler form α (or at least the existence of a Kähler form). Nevertheless, with the help of Theorem 6.6, we can show the following result.

Corollary 6.7. *Let X be a compact complex manifold in Fujiki's class $\mathcal{C}$. Then*

$$[\mathrm{Aut}_{[\alpha]}(X) : \mathrm{Aut}_0(X)] < \infty$$

for any big $(1,1)$-class $[\alpha]$ on X.

Note that Corollary 6.7 and Theorem 6.1 directly imply Corollary 6.4 with different approaches.

7. General compact complex manifolds and bimeromorphic groups

Let X be a (connected) compact complex manifold. Recall that we have the following exact sequence induced from the pullback restriction on the cohomology group

$$1 \to \mathrm{Aut}_\tau(X) \to \mathrm{Aut}(X) \to \mathrm{Aut}(X)|_{H^2(X,\mathbb{Q})} \to 1$$

where $\mathrm{Aut}_\tau(X)$ is defined as the kernel of the surjective map $\mathrm{Aut}(X) \to \mathrm{Aut}(X)|_{H^2(X,\mathbb{Q})}$. Recall that $\mathrm{Aut}(X)|_{H^2(X,\mathbb{Q})} \leqslant \mathrm{GL}(H^2(X,\mathbb{Q}))$ has bounded finite subgroups. So it suffices for us to focus on whether $\mathrm{Aut}_\tau(X)$ satisfies the Jordan property. Based on Corollary 6.7, we have the following question. Note that a positive answer towards Question 7.1 will directly imply the Jordan property for $\mathrm{Aut}(X)$.

Question 7.1. *Let X be a compact complex manifold. Is* $[\mathrm{Aut}_\tau(X) : \mathrm{Aut}_0(X)] < \infty$*?*

In non-algebraic cases, compact complex surfaces still behave well. Indeed, by the classification of surfaces, Prokhorov and Shramov [PS21, Theorems 1.6 and 1.7] showed that $\mathrm{Aut}(X)$ (and even the group $\mathrm{Bir}(X)$) are Jordan for any non-projective compact complex surface X. Nevertheless, Question 7.1 remains unknown even for surfaces.

Theorem 7.2 (cf. [PS21, Theorems 1.6 and 1.7])**.** *Let X be a connected compact complex surface. Then* $\mathrm{Bir}(X)$ *is not Jordan if and only if X is bimeromorphic to $E \times \mathbb{P}^1$, where E is an elliptic curve. Moreover, there always exists a constant $R = R(X)$ such that every finite subgroup of* $\mathrm{Bir}(X)$ *is generated by at most R elements.*

In dimension three, Prokhorov and Shramov [PS22] showed the Jordan property for $\mathrm{Bir}(X)$ when X is a non-uniruled Kähler threefold with non-zero irregularity. In higher dimensions with large Kodaira dimension, Loginov [Log25, Theorem 1.2] verified the Jordan property for the bimeromorphic automorphism group. The proof is based on the study of the bounded finite subgroup property for the pluricanonical representation.

Theorem 7.3 (cf. [Log25, Theorem 1.2])**.** *Let X be an n-dimensional compact complex variety with the Kodaira dimension $\kappa(X) \geqslant n-2$. Then the group* $\mathrm{Bir}(X)$ *is Jordan.*

Jia introduced the notion of T-Jordan property.

Definition 7.4. ([Jia23, Definition 1.2]) A group G is called *T-Jordan* (alternatively, we say that G has the *T-Jordan property*) if there is a constant $J(G)$ such that every torsion subgroup H of G has an abelian subgroup H_1 with the index $[H : H_1] \leqslant J(G)$.

Lee [Lee76] proved the T-Jordan property for connected real Lie groups.

Theorem 7.5. ([Lee76]) *A connected real Lie group is T-Jordan.*

Jia asked the following question:

Question 7.6 (cf. [Jia23, Conjecture 1.3])**.** *Let X be a compact complex manifold. Does* $\mathrm{Aut}(X)$ *satisfy the T-Jordan property?*

Jia and Meng confirmed the T-Jordan property for Fujiki's class $\mathcal{C}$ (cf. [JM24, Corollary 1.5]).

We introduce the following notations as in [Jia23]: Let Ξ be the set of smooth compact complex surfaces X in class VII with the algebraic dimension $a(X) = 0$ and the second Betti number $b_2(X) > 0$. Let $\Xi_0 \subseteq \Xi$ be those surfaces which have no curve. Note that $\Xi_0 = \varnothing$ if the Global Spherical Shell conjecture is true and it is known when $b_2(X) = 1$ by Teleman [Tel05].

Proposition 7.7 ([Jia23, Proposition 1.5]). *Let X be a compact complex surface which is not in Ξ_0. Then $\mathrm{Aut}(X)$ is T-Jordan.*

Theorem 7.8 ([Jia23, Theorem 1.7]). *Let X be a smooth compact complex surface. Then any torsion subgroup $G \leqslant \mathrm{Aut}(X)$ admits an abelian subgroup H with finite index $I = [G : H]$.*

Remark 7.9. When $X \in \Xi_0$, the index I in Theorem 7.8 depends on the group G (not just on the surface X).

Note that under settings of non-compact (connected) complex manifolds, or diffeomorphism groups of compact Riemannian manifolds, the Jordan property no longer holds in general. We refer to [CPS14], [Pop15], [Pop18] and [Zar19] for the counter examples; see also [BZ17], [BZ21], [MiRT15], [MiR19] and [MiR20] for positive cases. Here, we highlight a general result by Mundet i Riera.

Theorem 7.10 ([MiR19, Theorem 1.2]). *Let X be a compact real manifold with non-zero Euler characteristic. Then the group of (C^∞-)diffeomorphisms of X is Jordan.*

Again, we refer to Mundet i Riera [MiR19, §1] for the excellent survey on groups acting smoothly or continuously on smooth manifolds.

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